ELEC5307 Advanced Signal Processing with Deep Learning

Project 2 - Fruit Classification Challenge

Group 8	
Student Name	SID
Haoming Qu	540347724
Siyuan Zhuang	530388443
Jingyan Zhang	540891065

0. Contribution

Haoming Qu was responsible for drafting the *Methodology* and *Related Work* sections of the report, designing the attention module, conducting preliminary experiments, and implementing the core code. (40%)

Siyuan Zhuang contributed to the *Introduction* section and parts of the experimental analysis and conducted experiments involving different activation functions, optimizers, and learning-rate schedulers. (30%)

Jingyan Zhang completed the *Conclusion* section and portions of the experimental analysis, and designed and evaluated experiments comparing various data augmentation strategies and model architectures. (30%)

1. Introduction

Fruit recognition supports agricultural sorting, smart retail, and quality inspection. In real data, images often have low contrast, strong deformation, cluttered backgrounds, and occlusions. These issues reduce a network's sensitivity to key boundaries and fine-grained textures, which leads to confusion between similar classes. To improve robustness, we use a lightweight ResNet-18 backbone and add a Synergistic Channel–Spatial–Pixel Attention (SCSPA) module. The idea is to coordinate attention across three scales in the order "semantics - structure - details." Channel re-calibration suppresses task-irrelevant channels and amplifies useful ones. Multi-scale spatial convolution then aggregates structural cues. A final pixel-level saliency mask enhances micro features such as surface spots and deformation edges. Together, these steps form a simple three-stage, collaborative mechanism.

In our design, the attention block is placed at the end of the residual branch in every BasicBlock of stages 3 and 4 (256/512 channels). It sits after Conv2 and before the addition with the identity. This position keeps the extra computation low and improves the capture of discriminative regions. We also include a pixel–channel coupling to strengthen weak-texture areas further. Experiments show that adding SCSPA increases validation accuracy to 99.65%, which is a 0.35% gain over the baseline without SCSPA, while the inference speed remains essentially the same as ResNet-18.

2. Related works

Residual learning makes deep networks easier to train by adding identity skip connections. These connections can reduce training problems and help avoid gradient vanishing (He et al., 2016). As the network becomes deeper, residual connections can indirectly expand the receptive field while maintaining stable gradient flow. However, in complex scenarios, the network may perform well in terms of semantics but poorly in details (deep networks enhance the grasp of relevant regions or objects as a whole, but ignore the details and texture information at the shallow layers). When the image contrast is low, the shape undergoes significant changes, or parts are occluded, higher-level features are helpful for the overall picture, but fine edges and soft textures are smoothed. This leads to confusion of similar categories. The attention mechanism works by assigning more weight to useful features at the channel, spatial, and pixel levels (Woo et al., 2018; Sunil & Jaidhar, 2023).

These issues are common in agricultural vision and fruit recognition. Low light and similar backgrounds reduce contrast. Stacking, occlusion, and different ripeness levels change appearance and shape. Simple backbone networks using only higher-level features may not be sufficient to handle fine details. At the same time, transfer learning is a practical solution. Using pre-trained models on ImageNet or other public datasets provides powerful edge, texture, and shape features and makes the training of small-scale tasks more stable. For example, Behra et al. (2021) achieved an accuracy of 99.8% for papaya recognition using a transfer-learned ResNet-18 model and fewer training rounds. Shahi et al. (2022) used a MobileNetV2 model with channel attention mechanism to classify fruit categories and achieved an accuracy of over 95%, 4.3% higher than the model without the attention mechanism. Lokesh and Sentil (2025) used a triple attention mechanism combined with depthwise separable convolution to handle the frost damage of pomegranates, achieving an accuracy of

99.14% and a recall rate of 97.2%. These results show that transfer learning and attention mechanisms can handle noise and interference in the agricultural vision recognition field and improve training effectiveness.

Compared with previous work, we have adopted a joint channel-spatial-pixel attention design for fruit recognition and used a ResNet-18 model with transfer learning as the backbone. Through the design of the attention model, it can enhance fine edges and weak textures while maintaining speed close to the basic model. This method improves robustness in low contrast, shape changes, and interference from different categories.

3. Methodology

As noted earlier, residual networks are standard backbones in modern vision systems, with ResNet-18/34 among the most widely used variants. Both architectures employ a BasicBlock composed of two 3×3 convolutions and build hierarchical representations using [2, 2, 2, 2] blocks for ResNet-18 and [3, 4, 6, 3] for ResNet-34. In each stage, the first residual block performs downsampling by setting stride = 2 in its first convolution; when the spatial resolution or channel dimension changes, the identity (skip) branch is aligned using a 1×1 convolution with the same stride. Following this design, we adopt the lightweight ResNet-18 backbone to optimize fruit classification. **Figure 1.** presents the original ResNet-18 architecture, and **Figure 2.** illustrates the attention-augmented ResNet-18 in which attention modules are inserted into key residual units.

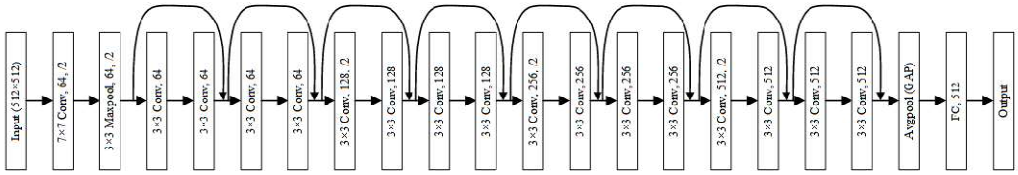


Figure 1. Schematic of the vanilla ResNet-18. Each BasicBlock contains two 3×3 convolutions and stages are stacked as [2, 2, 2, 2]; the first block of each stage downsamples with stride = 2.

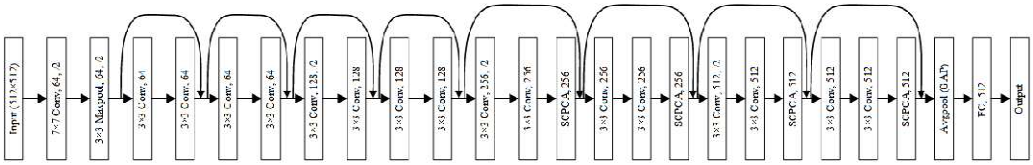


Figure 2. ResNet-18 with inserted attention modules.

Figure 2 places the SCSPA module at the end of the residual branch in every BasicBlock of ResNet-18 stages 3–4 (256/512 channels), adding minimal overhead while improving discrimination of fine textures and salient regions. SCSPA cascades Channel, Spatial, and Pixel attention in a single pipeline to reinforce global semantics, local structures, and fine-grained details. In the CA branch, global average/max pooling is passed through a shared two-layer 1×1 Conv-MLP to model inter-channel dependencies; the resulting weights re-scale features channel-wise, enhancing informative channels and suppressing redundant ones. The re-weighted features then enter the SA branch, which applies depthwise-separable convolutions with multi-scale kernels (5×5, 1×7, 1×11, 1×21); parallel outputs are summed and fused via a 1×1 convolution to produce a holistic spatial map. Next, the PA sub-module generates a pixel-level saliency mask using a lightweight two-layer Conv–Sigmoid head for precise localisation. Finally, SA and PA maps are multiplied and applied to the CA-enhanced features to yield the attention tensor, achieving joint optimisation across channel, spatial, and pixel dimensions, as shown in **Figure 3**.

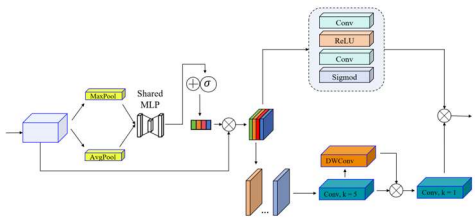


Figure 3. Structure of SCSPA module.

As illustrated in **Figure 3**, the SCSPA module integrates channel, spatial, and pixel attention in a cascaded manner. The left branch models inter-channel dependencies, the bottom branch captures multi-scale spatial relations, and the top branch refines pixel-level saliency. Their outputs are multiplicatively fused to form the final attention tensor.

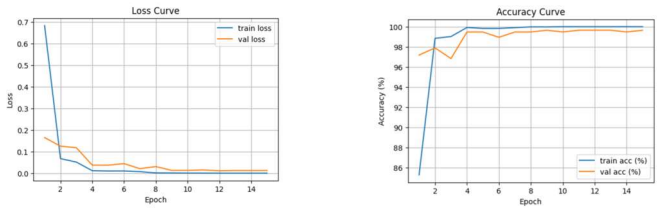
4. Experiment Design and Analysis

In our experimental design, we adopted the AdamW optimizer with an initial learning rate of $3 \times 10^{-4}$and a weight decay of 1×10^{-4}. We employed a cosine annealing learning-rate scheduler (CosineAnnealingLR) over 15 epochs without restarts, i.e., a single-cycle cosine anneal. The scheduling rule is given in **Formula. 1**.

$$lr(t) = \eta_{\min} + \frac{1}{2}\left(lr_0 - \eta_{\min}\right)\left(1 + \cos\left(\pi t / T_{\max}\right)\right), \quad t = 0,1,\ldots,15$$
$$where,\ lr_0 = 3\times10^{-4},\ \eta_{\min} = 0,\ T_{\max} = 15. \tag{1}$$

The dataset was split into training and validation sets with a 9:1 ratio. The original image size was 512×512. For the training set, we applied data augmentation using center cropping to 448×448 and normalization with ImageNet mean and standard deviation. For the validation set, we did not apply center cropping; we only used the same normalization. The batch size was 64, and four worker threads were used for data loading. The model was trained for 15 epochs with cross-entropy loss. To accelerate training, we used Google Colab with an L4 GPU (22.4 GB VRAM).

After 15 epochs, the model achieved 99.65% accuracy on the validation set. **Figure 4** shows the training curves: **Figure 4(a)** presents the loss, and **Figure 4(b)** shows the accuracy. During the first 2–4 epochs, accuracy increased rapidly while loss dropped sharply. Thereafter, both training and validation maintained high accuracy and near-zero loss, with a very small generalization gap and no signs of overfitting. Minor fluctuations in a few epochs are attributable to noise or small changes due to learning-rate annealing.



a) Loss curve b) Accuracy curve

Figure 4. Accuracy curve and loss curve

We further conducted ablation and comparative experiments to demonstrate the superiority of our model. Specifically, we evaluated different optimizers, learning-rate schedulers, data-augmentation strategies, activation functions, and backbone architectures. For clarity, the results are organized by category. Owing to space limitations, we report only the **best** validation accuracy achieved within each experimental group. To ensure consistent data partitioning and fair comparison, we fixed the random seed across all experiments.

4.1 Activation Function

In this section, we evaluate alternative activation functions—ELU, SiLU, and LeakyReLU. The results are summarized in **Table 1**.

Table 1. Result of different activation functions

Activation	Optimizer	Accuracy
LeakyReLU	AdamW	99.3%
SiLU	AdamW	98.24%
ELU	AdamW	97.18%
ReLU (Ours)	**AdamW**	**99.65%**

From Table 1, the model using **ReLU** achieves the highest validation accuracy. In contrast, models with **SiLU**, **ELU**, or **LeakyReLU** perform worse. A plausible explanation is that, although these smoother activations can provide better gradient flow or avoid dead neurons in theory, they reduce activation sparsity and shift the activation distribution, which may weaken implicit regularization and thus hurt generalization in our setting.

4.2 Optimizer

In this section, we compare optimizers including Adam, SGD, and SGD with momentum; the results are reported in **Table 2.**

Table 2. Result of different optimizers

Activation	Optimizer	Accuracy
Relu	Adam	99.47%
Relu	SGD + Nesterov Momentum	99.3%
Relu	SGD	99.12%
ReLU (Ours)	**AdamW (Ours)**	**99.65%**

As shown in **Table 2**, AdamW attains the best validation accuracy (99.65%) for our configuration, outperforming Adam and the two SGD variants. We attribute this to AdamW's adaptive preconditioning and decoupled weight decay, which stabilize training under our gradient scale. Note that we used a relatively simple schedule for SGD here; some work indicates that with carefully tuned schedules (e.g., warm restarts) and longer training, SGD often closes the gap (Ruder, 2016).

4.3 Data Augmentation

We evaluated several augmentation schemes; detailed settings and results are reported in **Table 3**.

Table 3. Result of different data augmentations

Augmentation	Size	Accuracy
CenterCrop	224	94.37%
CenterCrop	336	98.77%
RandomCrop	224	99.47%
RandomCrop	336	99.3%
RandomCrop	448	99.47%
None	512	99.47%
CenterCrop (Ours)	448	**99.65%**

Our results show that crop size clearly affects accuracy. When we use CenterCrop with sizes 224, 336, and 448, the accuracy rate shows a monotonically increasing trend. This means small crops will lose important details, while larger crops keep more details of the object. RandomCrop gives results close to using no augmentation at all, no matter the size. Therefore, for this dataset, the extra benefit is not apparent. In contrast, CenterCrop-448 has higher accuracy than the experiment without the augmentation setting, which indicates it fits the test data better and gives the model the correct information to help it improve.

4.4 Learning Rate Scheduler

We evaluated several learning-rate schedulers; the detailed settings and results are summarized in **Table 4**. In the step-decay setting, the learning rate was reduced by half at the 4th and 12th epochs.

Table 4. Result of different LR Schedulers

LR Scheduler	Optimizer	Accuracy
Step	AdamW	99.47%
OneCycleLR	AdamW	99.3%
CosineAnnealingLR (Ours)	AdamW	**99.65%**

Under our task and hyperparameters, **CosineAnnealingLR** (no warm-up, single smooth cycle) delivers better performance and generalization than both **OneCycleLR** and **step-based** schedules.

4.5 Model Comparison

We compare four representative backbones—ResNet-18, ResNet-34, VGG-16, and Swin-Transformer-Tiny. All the models adopted the same training scheme, and the detailed results are shown in Table 5.

Table 5. Result of different models

Model	Attention	Accuracy	Training speed (s/epoch)
Resnet-18	None	99.3%	**~26.1s**
Resnet-34	None	99.47%	~39.1s
VGG-16	None	97.54%	~42.2s
Swin-transformer-tiny	self-attention	**99.82%**	~39.4s
Resnet-18 (Ours)	SCSPA	99.65%	~27.6s

According to the data in Table 5, the validation accuracy of Swin-Transformer-Tiny is the highest (99.82%), but each training round takes approximately 39.4 seconds. In contrast, our ResNet-18 + SCSPA achieves an accuracy of 99.65% and reduces the training time to approximately 27.6 seconds per round, achieving a better balance between accuracy and efficiency. In traditional CNN benchmark models, ResNet-34 has a slightly improved accuracy of 99.47%, but the training time increases to approximately 39.1 seconds per round; VGG-16 has the slowest training speed (about 42.2 seconds per round) and the lowest accuracy (97.54%). Overall, when considering both accuracy and computational cost, ResNet-18 + SCSPA is the better choice.

5. Conclusion

To summarize, we have proposed a lightweight fruit classification model that builds upon ResNet-18 and incorporates a collaborative channel-spatial-pixel attention (SCSPA) module. Under the same 9:1 split and training scheme, our model achieves an accuracy of 99.65% on the validation set, which is close to 99.82% of Swin-Tiny. Moreover, the training speed of our model is approximately 40% faster per epoch than ResNet-34 and Swin-Transformer-Tiny, achieving a better balance between accuracy and efficiency. Abandonment studies and comparative experiments indicate that the generalization ability of “AdamW + CosineAnnealingLR” is superior to other settings. ReLU outperforms smooth activation functions in this small dataset and short cycle scenario, and CenterCrop-448 has the best match with the test distribution among the enhancement options.

The current results are based on a single dataset primarily containing one object. To enhance robustness against clutter and occlusion, future work will consider more diverse enhancement methods (color/lighting variations, minor geometric distortions) and evaluate on more complex datasets. Meanwhile, this project can be expanded to actual fruit target detection tasks to improve practical application.

Reference:

Behera, S. K., Rath, A. K., & Sethy, P. K. (2021). Maturity status classification of papaya fruits based on machine learning and transfer learning approach. *Information Processing in Agriculture*, *8*(2), 244-250.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

Lokesh, C. K., & Senthil, S. (2025). A Novel Triple Attention-Based Deep Learning Framework for Accurate Pomegranate Disease Detection. *Journal of Phytopathology*, *173*(4), e70118.

Ruder, S. (2016). An overview of gradient descent optimization algorithms. *arXiv preprint arXiv:1609.04747*.

Shahi, T. B., Sitaula, C., Neupane, A., & Guo, W. (2022). Fruit classification using attention-based MobileNetV2 for industrial applications. *Plos one*, *17*(2), e0264586.

Sunil, C. K., & Jaidhar, C. D. (2023). Tomato plant disease classification using multilevel feature fusion with adaptive channel spatial and pixel attention mechanism. *Expert Systems with Applications*, *228*, 120381.

Woo, S., Park, J., Lee, J. Y., & Kweon, I. S. (2018). Cbam: Convolutional block attention module. In *Proceedings of the European conference on computer vision (ECCV)* (pp. 3-19).